

Naive Bayes Mathematical Solution

Dataset:

P1: Fever=1, Cough=0, Disease=0

P2: Fever=1, Cough=1, Disease=1

P3: Fever=0, Cough=1, Disease=0

New Patient: Fever=0, Cough=0

1. Prior Probabilities

$P(D=1)=1/3$, $P(D=0)=2/3$

2. Likelihoods with Laplace Smoothing

Formula: $P(x=v|D=d) = (\text{count}(x=v, D=d)+1)/(\text{count}(D=d)+2)$

Disease = 1

$P(\text{Fever}=0|D=1)=(0+1)/(1+2)=1/3$

$P(\text{Cough}=0|D=1)=(0+1)/(1+2)=1/3$

$P(X|D=1)=1/3 \times 1/3 = 1/9$

Posterior Numerator = $(1/9)(1/3)=1/27$

Disease = 0

$P(\text{Fever}=0|D=0)=(1+1)/(2+2)=1/2$

$P(\text{Cough}=0|D=0)=(1+1)/(2+2)=1/2$

$P(X|D=0)=1/2 \times 1/2 = 1/4$

Posterior Numerator = $(1/4)(2/3)=1/6$

3. Normalization

$P(X)=1/27 + 1/6 = 11/54$

$P(D=1|X)=(1/27)/(11/54)=2/11\approx 0.1818$

$P(D=0|X)=(1/6)/(11/54)=9/11\approx 0.8182$

Final Prediction

Since $0.8182 > 0.1818$, Prediction = No Disease